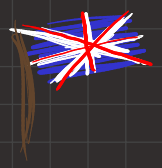
DISTRIBUTED SYSTEMS NOTES



Carlo Da Rome NOV. 5. 2026

CH. 1 G.P.E.

QUESTION 1: WHAT IS A DISTRIBUTED SYSTEM?

ANSWER: A DISTRIBUTED SYSTEM IS A SET OF COMPUTERS THAT COOPERATE TO EXECUTE OPERATIONS OR TO STORE DATAS, THEY DON'T USE CLOCK FOR SINCRONIZATION, THE SINCRONIZATION IS POSSIBLE THROUGH MESSAGES.

- A LOT OF PROBLEMS IN THIS TYPE OF SYSTEM CAN BE TRACED BACK TO G.P.E. (global predicate evaluation) THAT IS EVALUATE IF THE SYSTEM SATISFIES A CONDITION.
- AN ASINCRONOS DYSTRIBUTED SYSTEM (A.D.S.) IS A SYSTEM WITH THE FOLLOWING PROPERTIES:

1. THE M PROCESSES IN THE SYSTEM IS REPRESENTED WITH THE SET $(P_1, \dots, P_m)$.
2. THE COMMUNICATION CHANNELS IS BASICALLY NOT FIFO, THIS MEANS THAT THE ORDER OF SENDED MESSAGES COULD BE DIFFERENT TO THE RECIEVE ORDER.
3. PROCESSES ARE STRONG CONNECTED, THIS MEANS THAT PROCESSES ARE ALL CONNECTED EACH OTHER.
4. CHANNELS ARE RELIABLE... WE CAN ASSUME THAT WE CANNOT HAVE DATA LOST OR COMPROMIZED.

5. SINCRONIZATION PASSES ONLY THROUGH MESSAGES

THIS IS A THEORETICAL MODEL, IT DOEST EXIST IN REALITY 'CAUSE IS NOT POSSIBLE TO REALIZE THAT BUT IT IS A GOOD ABSTRACTION BECAUSE.

1. IS VERY SIMPLE AND DOESN'T HAVE ANY BOUND IN TERM OF SPEED OR RETARD, IF ONE SOLUTION IS GOOD ON THIS MODEL IT IS ALSO GOOD IN REAL LIFE.

2. SOME LAYERS IN REAL SYSTEMS FOLLOW THIS MODEL (RULE 4 IS IMPOSSIBLE TO SATISFIES IN REAL LIFE)

- IN A DISTRIBUITED COMPUTATION WE CAN HAVE 3 TYPE OF EVENTS

1. INTERNAL EVENT.

IT ONLY CHANGE THE STATE OF THE PROCESS THAT EXECUTES THE EVENT.

2. EXTERNAL EVENT.

2.1 SEND(m)

2.2 RECIEVE(m)

THESE TWO TYPE OF EVENTS ATTACH A MESSAGE INDICATED WITH THE LEETER "m"

CH.2 GLOBAL STATES, CUT AND CONSISTENCY

QUESTION 2: WHAT IS CONSISTENCY?

ANSWER: IN A D.S. WE HAVE CONSISTENCY WHEN THE MONITOR PROCESS p_0 (PROCESS THAT MAKE G.P.C.) RECIEVE COMPATIBLE STATES.

- HISTORY.

IN DISTRIBUITED SYSTEM WE HAVE TWO TYPE OF

HISTORY...

1. LOCAL HISTORY.

IS THE SET OF THE EVENTS EXECUTED BY A SPECIFIC PROCESS P_i

$h_i = e_i^1 e_i^2 e_i^3 \dots$ POTENCIALLY INFINITE...

↑ WHERE i IS REFERRED TO THE PROCESS P_i

2. GLOBAL HISTORY.

IS THE SET OF ALL THE EVENTS EXECUTED BY ALL THE PROCESSES IN THE DISTRIBUTED SYSTEM.

$H = h_1 \cup h_2 \cup h_3 \cup \dots \cup h_m$

$H = e_1^1 \dots e_1^{\dots} \cup e_2^1 \dots e_2^{\dots} \cup [\dots]$

- CAUSAL PRECEDENCE.

THE CAUSAL PRECEDENCE "$\rightarrow$" IS A RELATION ON THE EVENTS IN H

IF $e \rightarrow e' \Rightarrow e$ INFLUENCE e' AND FOLLOWS THE FOLLOWING PROPERTIES:

1. IF e, e' IS EVENT OF THE SAME PROCESS P_i AND e PRECEDES e' IN THE LOCAL HISTORY h_i THEN $e \rightarrow e'$.
2. IF e = SEND(m) AND e' = RECIEVE(m) THEN $e \rightarrow e'$
3. IF $e \rightarrow e'$ AND $e' \rightarrow e''$ THEN $e \rightarrow e''$.

- GLOBAL STATES, CUT, RUN

- TO PERFORM A G.P.E. THE MONITOR p_0 REQUIRES THE GLOBAL STATE OF THE SISTEM... THE GLOBAL STATE Σ IS DEFINED AS $\Sigma = \sigma_1, \sigma_2, \dots, \sigma_m$ WHERE σ_i IS THE LOCAL STATE OF P_i.
- A CUT IS A SUBSET IN THE GLOBAL HISTORY.

$C = h_1^{e_1} \cup \dots h_m^{e_m}$.

THE TUPLE $(e_1,\dots,e_m)$ INDICATE THE LAST EVENT TO CONSIDER IN THE LOCAL HISTORY.

- A RUN IS A TOTAL ORDER ON H, IF e PRECEDES e' IN LOCAL HISTORY h_i OF p_i THEN e PRECEDES e' IN R.
- A CUT IS CONSISTENT IF LEFT CLOSED OR RATHER: IF $\forall e, e' \in H$ AND $e' \in C$ AND $e \to e' \Rightarrow e \in C$.
- A RUN IS CONSISTENT IF $\forall e, e'$ $e \to e' \Rightarrow$ e APPEARS BEFORE e' IN R.

CH. 3 MONITORING A D.S.

TO MONITOR A D.S. WE CAN APPLY TWO TYPE OF PREDICATE... ACTIVE MONITOR OR PASSIVE MONITOR.

ACTIVE MONITOR.

IN SYSTEMS WITH ACTIVE MONITOR, THE MONITOR PROCESS p_0 ASKS CONTINOUSLY TO THE OTHER PROCESSES THEIR LOCAL STATE. WE CAN NOTICE SOME PROBLEMS IN THIS TYPE OF SYSTEM.

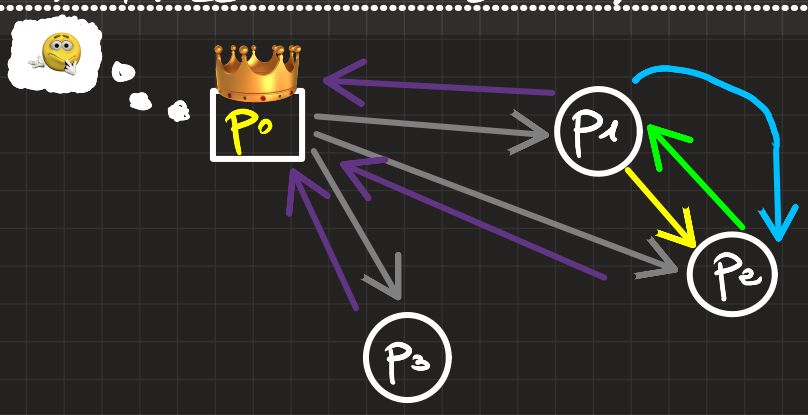
1. GLOBAL STATES THAT THE MONITOR RETRIEVE CAN BE INCONCOSISTENT AND ENCREASING THE FREQUENCY OF THE REQUESTS THE PROBLEM IS NOT FIXED.

2. GHOST DEADLOCK

WE ALSO COULD HAVE GHOST DEADLOCK...

BASIC EXAMPLE OF GHOST DEADLOCK.

EXAMPLE 1.



IN THE EXAMPLE WHE HAVE THE FOLLOWING SITUATION:

P_1 SEND TO P_2 A MESSAGE AND IT WAIT FOR A CONFIRMATION MESSAGE FROM P_2

P_2 SEND TO P_1 THE CONFIRMATION MESSAGE ... AND WAIT FOR A CONFIRMATION MESSAGE FROM P_1 ...

IF THE MONITOR P_0 SENDS THE STATE REQUEST IT CAN ARRIVES TO P_1 AND P_2 IN THE FOLLOWING MOMENT...

P_1 : WAITING FOR P_2 CONFIRMATION MESSAGE
P_2 : WAITING FOR P_1 CONFIRMATION MESSAGE

(THIS IS CAUSED BY THE IMPREVEDIBLE RETARDS ON THE COMMUNICATION LINE)

THE GLOBAL STATE RETRIVED IS AN INCONSISTENT STATE AND IT CONTAINS A GHOST DEADLOCK, OR RATHER A DEADLOCK THAT DOESN'T EXIST IN THE SYSTEM BUT EXIST IN G.P.E..

- PASSIVE MONITOR.

IN SYSTEM WITH PASSIVE MONITOR THE MONITOR PROCESS P_0 DOESN'T ASK TO THE OTHER THEIR STATE BUT THE PROCESSES IN D.S. SENDS TO THE MONITOR ONLY RELEVANT INFORMATIONS FOR THE G.P.E.

TO AVOID UNCONSISTENT CUT WE INTRODUCE FIFO RULE

- FIFO RULE

EVERY PROCESS IN THE DISTRIBUITED SYSTEM HAS 2 LEVEL 1) DELIVERY LEVEL 2) APPLICATION

LEVEL.
ALL THE MESSAGES ARRIVE IN THE DELIVERY LEVEL AND DELIVERED TO THE APPLICATION LEVEL FOLLOWING THE ORDER DESCRIBED BY THE **FIFO RULE** OR RATHER:

A COMMUNICATION BETWEEN p_i AND p_j FOLLOW THE **FIFO DELIVERY RULE** IF $\forall m, m'$

$$SEND_i(m) \rightarrow SEND_i(m') \Rightarrow DELIVER_j(m) \rightarrow DELIVER_j(m')$$

where the pedice indicates the process that execute the event.

CH. 4 CLOCK

- UNTIL NOW WE HAVE ANALIZED DISTRIBUITED SYSTEM WITH NO CLOCK, NOW, SUPPOSE TO HAVE A SYSTEM WHERE THE MONITOR RECIEVES ALL THE MESSAGES EXCHANGED BETWEEN THE PROCESSES WITH AN ATTACHED **TIMESTAMP** AND P_0 STORE THESE MESSAGES IN A QUEUE AND DELIVERS THESE PERIODICALLY TO THE APPLICATION LEVEL.
THIS TYPE OF SYSTEMS SATISFIES **CLOCK CONDITION**

- **CLOCK CONDITION**
A FUNCTION T THAT ASSOCIATES A TIMESTAMP TO AN EVENT SATISFIES THE CLOCK RULE IF $e \rightarrow e' \Rightarrow T(e) < T(e')$.
BUT THE CLOCK RULE CAN BE SATISFIED BY OTHER TYPE OF SYSTEMS...

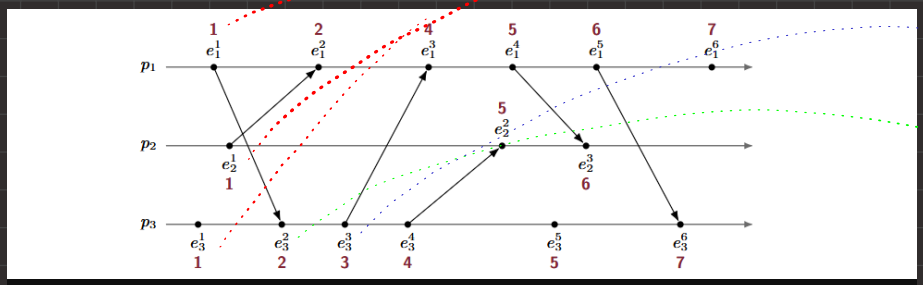
- LAND PORT LOGICAL CLOCKS.
SUPPOSE TO HAVE A COUNTER "LC" FOR EACH PROCESS p_i IN A DISTRIBUITED SYSTEM.
WHEN AN EVENT OCCOURRES THE LC ASSOCIATED TO THE EVENT FOLLOW THE FOLLOWING RULE:

$$LC(e_i) = \begin{cases} LC += 1 & \leftarrow \text{INTERNAL / SEND EVENT} \\ \max\{LC, TS(m)\} + 1 & \leftarrow \text{RECEIVE EVENT} \end{cases}$$

THIS TYPE OF SYSTEM ALSO SATISFIES THE **CLOCK CONDITION**...

TO BETTER COMPREND FOLLOW THE FOLLOWING EXAMPLE...

EXAMPLE 3.



- The first events e_1^1, e_1^2, and e_1^3 are either send events or internal events, so their clock values are 1. The requests sent in e_1^1 and e_1^2 carry a timestamp of 1.
- Event e_2^1 receives the request from process p_2 with timestamp 1: $\max\{1, 1\} + 1 = 2$. Similarly, e_3^3, which receives the request from p_1, gets a value of 2.
- Process p_3 sends a response, $LC(e_3^3) = 3$, and then sends a request to p_2, $LC(e_4^3) = 4$.
- Event e_3^1 receives the response from p_3 with timestamp 3: $\max\{2, 3\} + 1 = 4$. The clock of process p_1 jumps from 2 to 4 to catch up with p_3's past.
- Event e_2^2 receives the request from p_3 with timestamp 4: $\max\{1, 4\} + 1 = 5$.
- Process p_1 proceeds with the response to p_2, $LC(e_4^1) = 5$, the request to p_3, $LC(e_5^1) = 6$, and the internal event e_6^1, which has a value of 7.
- Event e_3^2 receives the response from p_1 with timestamp 5: $\max\{5, 5\} + 1 = 6$. Event e_5^3 is internal and has a value of 5; event e_6^3 receives the request sent in e_5^1 with timestamp 6: $\max\{5, 6\} + 1 = 7$. The clock of process p_3 jumps from 5 to 7 and never assumes the value 6.

- THESE TWO TYPE OF SYSTEM DON'T HAVE GAP DETECTION BECAUSE THEY CANNOT RECOGNIZE IF AN EVENT WITH LOWER TIMESTAMP THAN THE HIGHEST TIMESTAMP THAT THEY HAVE IN THEIR HAND CAN ARRIVE...

QUESTION 3: WHAT IS GAP DETECTION?

ANSWER: A SYSTEM HAS GAP DETECTION IF GIVEN $LC(e) < LC(e')$ CAN RECOGNIZE IF EXIST e'' WITH $LC(e) < LC(e'') < LC(e')$.

- HOW TO ADD GAP-DETECTION TO OUR SYSTEM?

WE CAN IMPLEMENT THIS IN A VERY SIMPLE WAY.

1) ADD FIFO CHANNELS IN EACH PROCESS OF THE D.D.

2) p_0 DELIVERS A MESSAGE ONLY IF IS STABLE.

- STABLE MESSAGE.

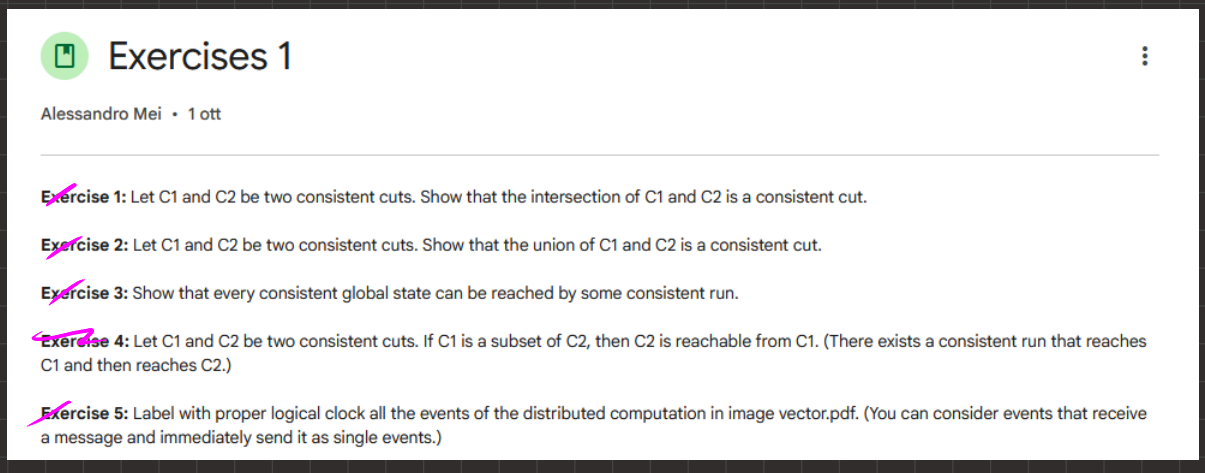
A MESSAGE m WITH TIMESTAMP $TS(m)$ IS STABLE FOR p_0 IF IT RECIEVE AT LEAST A MESSAGE m_1 WITH TIME STAMP $TS(m_1) > TS(m)$ FROM EACH PROCESS IN THE DISTRIBUITED SYSTEM.

WHY IT WORKS ???

BECAUSE IF p_0 RECIEVE A MESSAGE m WITH TIMESTAMP 4 FROM p_1, A MESSAGE m_1 WITH TIMESTAMP 5 FROM p_2 AND A MESSAGE m_{11} WITH TIMESTAMP 6 FROM p_3 IN A D.D. $\{p_0, p_1, p_2, p_3\}$, p_0 IS SURE TO NOT RECIEVE A MESSAGE WITH TIMESTAMP LOWER THAN 4 BECAUSE EACH PROCESS HAS A FIFO CHANNEL.

PROBLEM: A PROCESS CAN STOP SEND MESSAGES

SOLUTION: p_0 ASK TO STOPPED PROCESSES FOR CONFIRM MESSAGES, THESE RESPONSES WILL HAVE BIGGER TS THAN THE MESSAGES IN QUEUE.



Exercises 1

Alessandro Mei • 1 ott

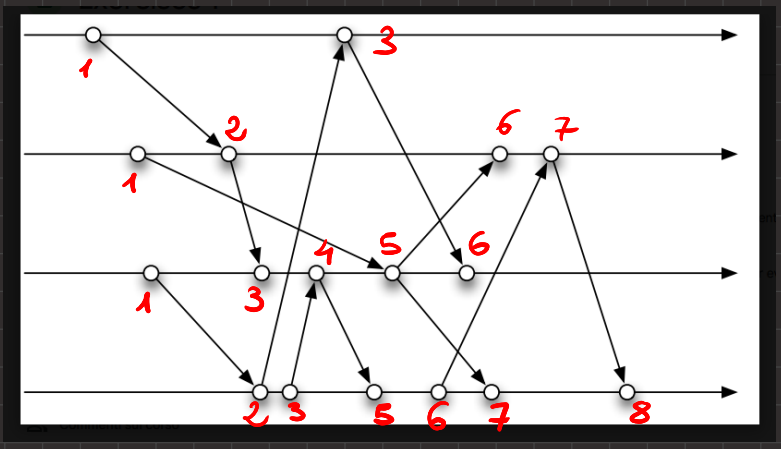
Exercise 1: Let C1 and C2 be two consistent cuts. Show that the intersection of C1 and C2 is a consistent cut.

Exercise 2: Let C1 and C2 be two consistent cuts. Show that the union of C1 and C2 is a consistent cut.

Exercise 3: Show that every consistent global state can be reached by some consistent run.

Exercise 4: Let C1 and C2 be two consistent cuts. If C1 is a subset of C2, then C2 is reachable from C1. (There exists a consistent run that reaches C1 and then reaches C2.)

Exercise 5: Label with proper logical clock all the events of the distributed computation in image vector.pdf. (You can consider events that receive a message and immediately send it as single events.)



vector.pdf

EXERCISE 1

LET e_1 AND e_2 CONSISTENT CUTS SHOW THAT THE INTERSECTION $(e_1 \cap e_2)$ IS A CONSISTEN CUT...

- A CUT IS CONSISTENT IF IS LEFT CLOSED, OR RATHER, IF $\forall e, e' \in H, e \in e \wedge e \to e' \Rightarrow e \in e$ WHERE "H" IS GLOBAL HISTORY AND "e" IS A CUT.

WE HAVE TO DEMONSTRATE THAT:

$\forall e, e' \in H, e' \in (e_1 \cap e_2), e \to e' \Rightarrow e \in (e_1 \cap e_2)$... AND THIS IS TRUE BECAUSE IF $e' \in (e_1 \cap e_2) \Rightarrow (e' \in e_1) \wedge (e' \in e_2)$

NOW, IF $e \to e'$ AND $e' \in (e_1 \cap e_2) \Rightarrow (e' \in e_1) \wedge (e' \in e_2) \Rightarrow$

$e \in e_1 \wedge e \in e_2 \Rightarrow e \in (e_1 \cap e_2)$ WHERE $e, e' \in H$.

EXERCISE 2

LET e_1 AND e_2 CONSISTENT CUTS SHOW THAT THE UNION $(e_1 \cup e_2)$ IS A CONSISTEN CUT...

- A CUT IS CONSISTENT IF...

WE HAVE TO DEMONSTRATE THAT:

$\forall e, e' \in H, e' \in (e_1 \cup e_2), e \to e' \Rightarrow e \in (e_1 \cup e_2)$... AND THIS IS TRUE BECAUSE IF $e' \in (e_1 \cup e_2) \Rightarrow (e' \in e_1) \vee (e' \in e_2)$

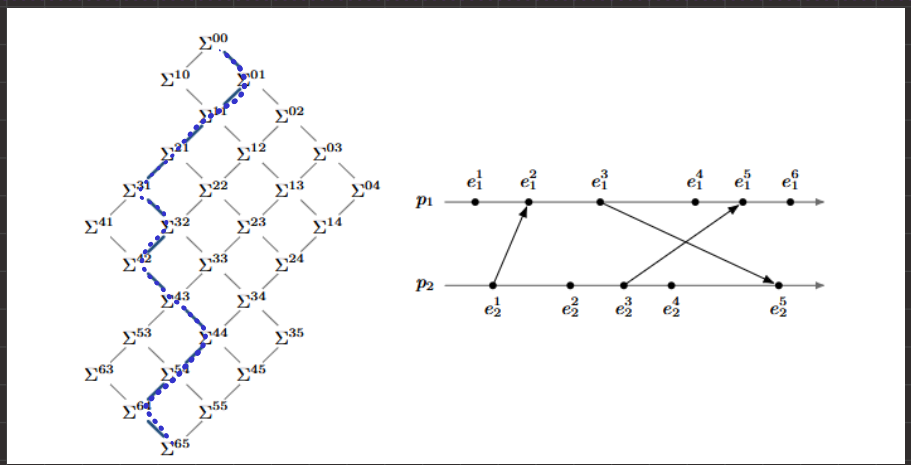
NOW, IF $e \to e'$ AND $e' \in (e_1 \cup e_2) \Rightarrow (e' \in e_1) \vee (e' \in e_2) \Rightarrow$

$e \in e_1 \vee e \in e_2 \Rightarrow e \in (e_1 \cup e_2)$ WHERE $e, e' \in H$.

EXERCISE 3

DEMONSTRATE THAT ALL THE CONSISTENT GLOBAL STATE CAN BE REACHED BY SOME CONSISTENT RUN...

- WE CAN REPRESENT A DISTRIBUTED COMPUTATION WITH THE LATTICE OF GLOBAL STATE, IN THIS REPRESENTATION WE HAVE STATES: $\Sigma^{00\ldots00}$, $\Sigma^{00\ldots01}$, $\Sigma^{00\ldots10}$ [...] AND THESE ARE ALL CONSISTENT STATES, THE NUMBERS ON TOP ARE THE NUMBER OF OPERATIONS EXECUTED BY THE i-TH PROCESS AND ARE m WHERE m IS THE NUMBER OF PROCESSES IN THE DISTRIBUITED SYSTEM
- A RUN IS A TOTAL ORDER ON A SUBSET OF THE HISTORY AND A RUN R IS CONSISTENT IF $\forall e, e' \in R$ IF $e \to e' \Rightarrow e$ APPEARS BEFORE e' IN R.
- IN A LATTICE OF GLOBAL STATES WE HAVE ALL THE POSSIBLE CONSISTENT STATES THAT A CERTAIN D.S. CAN REACH... IF WE MOVE FROM A STATE TO ANOTHER STATE THE NUMBERS ON THE STATES ARE DIFFERENT 'CAUSE SOME EVENT OCCOURRED AND THESE EVENT FORM A CONSISTENT RUN.



..... : $\Sigma^{00}\,\Sigma^{01}\,\Sigma^{11}\,\Sigma^{21}\,\Sigma^{31}\,\Sigma^{32}\,\Sigma^{42}\,\Sigma^{43}\,\Sigma^{44}\,\Sigma^{54}\,\Sigma^{64}\,\Sigma^{65}$

AND CORRESPONT TO

$e_2^1 e_1^1 e_1^2 e_1^3 e_2^2 e_1^4 e_2^3 e_2^4 e_1^5 e_1^6 e_2^5$.

THAT IS A CONSISTENT RUN

- WE CAN DEMONSTRATE IT BY INDUCTION...

BASE CASE :

WE CAN TAKE $\Sigma^{00...00}$, THAT IS THE INITIAL STATE THIS IS A CONSISTENT STATE AND THERE EXIST A RUN (NULL RUN) THAT REACH THIS STATE.

INDUCTIVE IP:

SUPPOSE VERIFIED UNTIL LEVEL $(m+1)$

INDUCTIVE CASE:

- NOW WE HAVE TO VERIFY IF IT IS REAL FOR LEVEL $(m+1)$
- WE HAVE SUPPOSED THAT UNTIL LEVEL $(m+1)$ WE HAVE CONSISTENT RUN FOR EACH STATE IN THE LATTICE, NOW TO GO TO A STATE IN LEVEL $(m+1)$ WE HAVE TO GENERATE AN EVENT AND WE HAVE 3 CASE OF EVENT

1) INTERNAL EVENT

   VERIFIED BECAUSE WILL EXIST A RUN WITH THIS EVENT

2) SEND //

3) RECIEVE //

∎

ESERCIZIO 4

LET c_1, c_2 CONSISTENT CUTS AND $c_1 \subseteq c_2$ THEN THERE EXIST A RUN THAT REACH c_1 AND THEN REACH c_2.

SUPPONING TO HAVE A CUT c_1 AND A CUT c_2

- IF $e' \in c_2 \wedge \forall e \in c_1 \; e \| e' \Rightarrow \exists R$ CONSISTENT RUN THAT REACH c_1 AND THEN c_2 BECAUSE IS ALWAYS

POSSIBLE TO ADD A NEW INDIPENDENT EVENT TO A RUN.

- IF $e' \in c_2 \wedge \exists e \in c_1 : e \to e' \Rightarrow \exists R$ CONSISTENT RUN THAT REACH c_1 AND THEN c_2 BECAUSE IS LEFT CLOSED

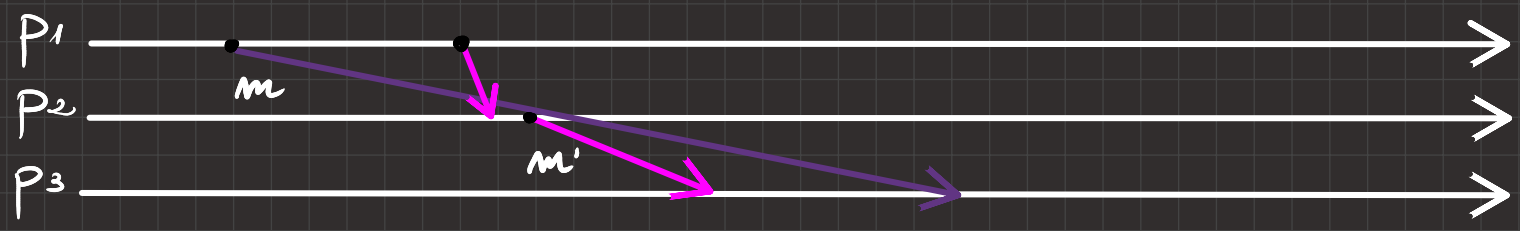
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CH.5 CAUSAL DELIVERY AND VECTOR CLOCKS.

Carlo Re Rome NOV. 6. 2026

IN THE MODELS ANALIZED UNTIL NOW WE CAN NOTICE A VERY STRONG PROBLEM

EXAMPLE 5.



WE CAN NOTICED THAT $SEND_i(m) \to SEND_i(m_1)$ BUT $DELIVER_j(m') \to DELIVER_j(m)$

BECAUSE THE SYSTEM ANALIZED CAN HANDLE VERY WELL MESSAGES FROM ONE SOURCHE BUT IT HAS PROBLEMS WITH MORE...

- CAUSAL DELIVERY RULE

A SYSTEM SUPPORTS CAUSAL DELIVERY RULE IF $\forall m, m'$ $SEND_i(m) \to SEND_j(m') \Rightarrow$ $DELIVER_k(m) \to DELIVER_k(m')$.

IF $i = j$ CAUSAL DELIVERY BECOME FIFO DELIVERY RULE

CAUSAL DELIVERY RULE IS A PERFECTIONATED

FIFO DELIVERY RULE.

- **STRONG CLOCK CONDITION**

A SYSTEM RESPECT THE STRONG CLOCK CONDITION IF GIVEN A FUNCTION T THAT ASSIGN A VALUE TO EACH EVENT $\forall e, e' \in H$ IF $e \rightarrow e' \Leftrightarrow T(e) < T(e')$.

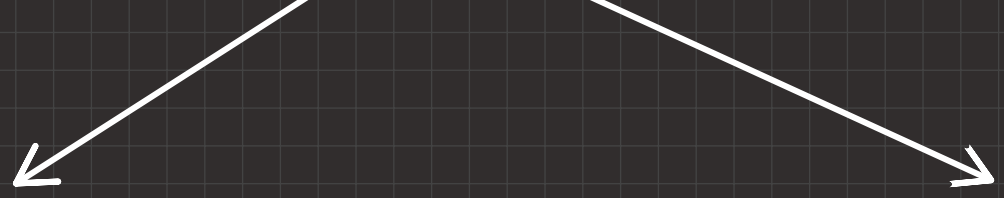
QUESTION 4. HOW TO IMPLEMENT THIS TWO IMPORTANT RULES IN A SYSTEM ??

ANSWER. VECTOR CLOCKS

- VECTOR CLOCKS.

PHASE 1) EACH PROCESS IN THE DISTRIBUTED SYSTEM INITIALIZE A VECTOR VC LONG IN CELLS AS THE SAME NUMBER OF PROCESSES IN THE D.S.

PHASE 2) IF AN EVENT OCCOURRES IN PROCESS p_i



INTERNAL / SEND

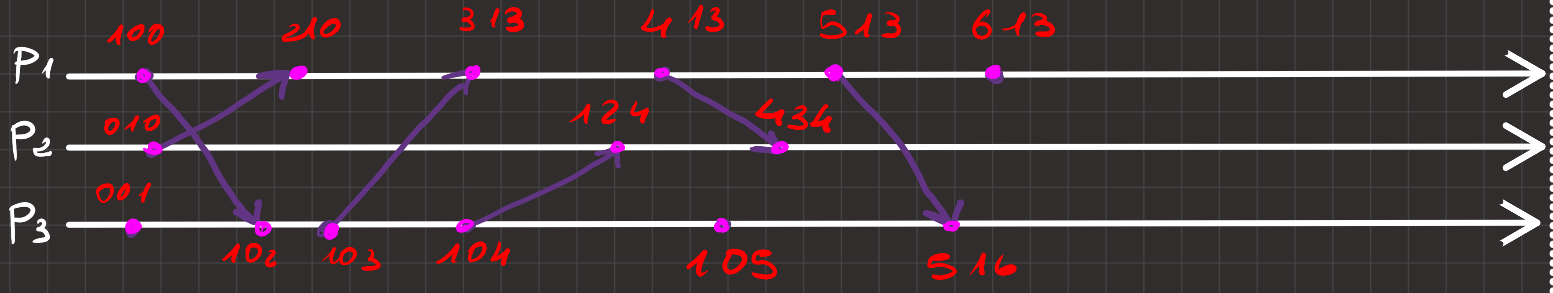
- $VC[i] += 1$

RECEIVING

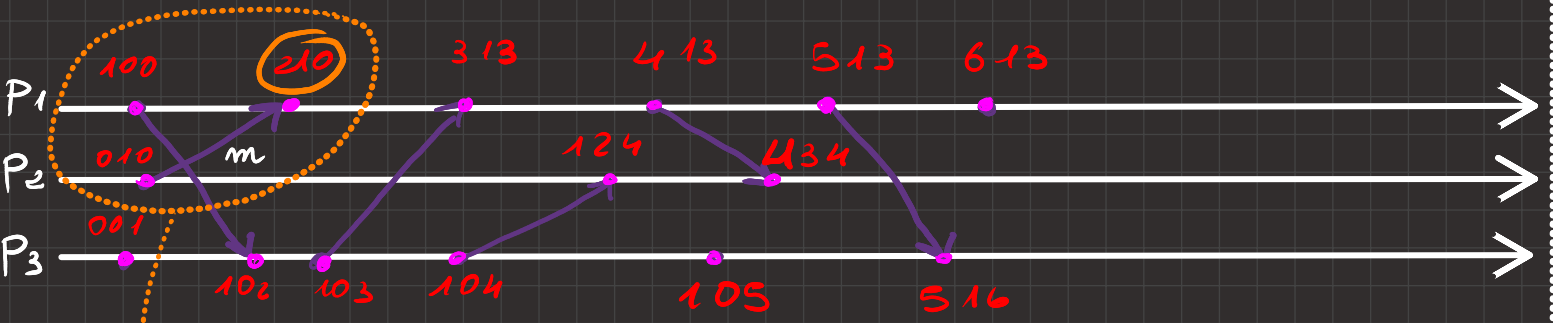
1. 1) $\forall_J \; VC[J] = max\{VC[J], TS(m)\}$
   2) $VC[i] += 1$
   (where J is a process.)
2. TS(m) is the VECTOR CLOCK ATTACHED TO THE MESSAGE m
3. EACH TIME THAT A PROCESS SENDS A MESSAGE IT ATTACH ITS VECTOR CLOCK TO THE MESSAGE (TS(m)).

EXAMPLE 3.

EXAMPLE OF A COMPUTATION



MORE IN DETAILS ...



210 BECAUSE
$TS(m) = 010$ WHILE
VC OF $p_1 = 100$

1) $VC[0] = \max\{1,0\} = 0$

$VC[1] = \max\{1,1\} = 0$

$VC[2] = \max\{0,0\} = 0$

2) $VC[i] += 1$

$VC = (2,1,0)$

BEFORE TO GO AHEAD WE HAVE TO TALK ABOUT

- CAUSAL HISTORY

THE CAUSAL HISTORY OF AN EVENT $\mathcal{O}(e)$ IS DEFINED AS $\mathcal{O}(e) = \{e' : e' \to e\} \cup \{e\}$

- COMPARE TWO CAUSAL VECTORS

TWO CAUSAL VECTOR ARE UNCOPARABLE IF

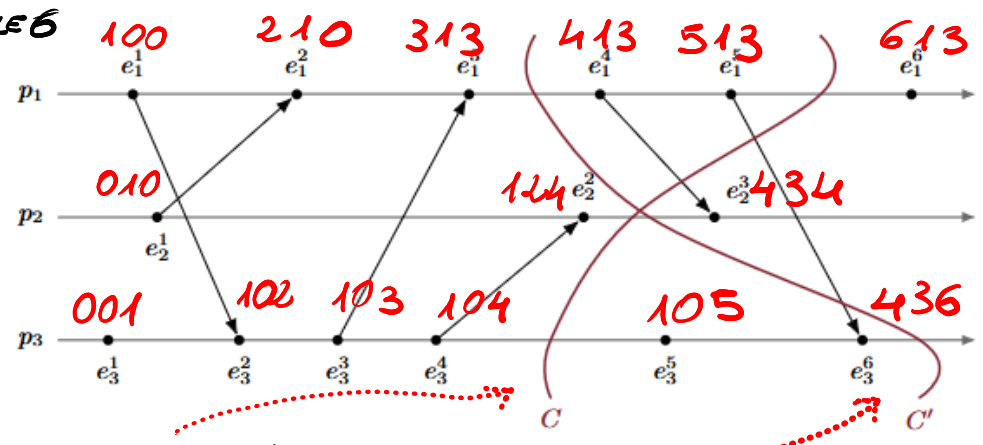
$!(CV(e) < CV(e')) \wedge !(CV(e) \geq CV(e'))$ ...

WHILE IF TWO CAUSAL VECTOR ARE COMPARABLE

$e \to e' \iff VC(e) < VC(e')$ (the system respect the clock condition)

- HOW TO KNOW IF A CUT IS CONSISTENT?

EXAMPLE6



$e = (5,2,4)$ $e' = (3,2,6)$

A CUT IS CONSINSTENT IF:

$VC(e_j^{e_j}) \leq (c_1, \dots, c_m)$ for each process p_j

WHERE $e_j^{e_j}$ IS THE LAST EVENT PERFORMED BY p_j IN THE CUT AND $(c_1, \dots, c_m)$ IS THE NUMBER OF EVENTS PERFORMED BY EACH PROCESS IN THE CUT

p_1 PERFORMED c_1 EVENTS IN THE CUT

p_2 PERFORMED c_2 EVENTS IN THE CUT

[...]

p_m PERFORMED c_m EVENTS IN THE CUT

THIS METHOD WORKS WELL BECAUSE IF A VC IS GRATHER THAN THE TUPLE, THIS MEANS THAT IT KNOW MORE EVENTS THAN THE EVENTS IN THE CUT AND IT IS COUNTING SOME EVENTS OUTSIDE THE CUT...

- SOMETHING MISSING IN THE MIDDLE?

THE SYSTEM USES THE WEAK GAP DETECTION.

$VC(e)[k]$ MEANS THAT e KNOW THE e EVENTS PERFORMED BY P_k ... IF

$VC(e)[k] < VC(e')[k] \Rightarrow \exists$ EVENT f PERFORMED BY p_k WHERE $f \not\to e \wedge f \to e'$.

THIS IS WEAK GAP DETECTION BECAUSE

"e DON'T KNOW ABOUT f" IS NOT "f CAME AFTER e"

FOR EXAMPLE: $VC(e_1^2) = (2,1,0)$ AND $VC(e_2^3) = (4,3,4)$

- CAUSAL DELIVERY WITH VECTOR CLOCK.

WE HAVE 3 ACTORS

- p_1, p_2. TWO COMMON PROCESSES
- p_0. THE MONITOR PROCESS

EACH MESSAGE m TO p_0 HAS $TS(m)$ ATTACHED

IF THE SENDER IS p_J:

* $TS(m)[J]$: INDICATE THE NUMBER OF NOTIFICATIONS SENDED BY p_J
* $TS(m)[K]$: INDICATE HOW MUCH NOTIFICATIONS p_J KNOW ABOUT SENDED BY p_K IN THE MOMENT WHEN p_J SEND THE CURRENT.

THE MONITOR HAS A VECTOR, D, WHERE $D[k]$ INDICATE HOW MUCH NOTIFICATIONS BY p_K THE MONITOR HAS ALREDY DELIVERED.

- DR3 RULE



AGENZIA DOGANE MONOPOLI

WHEN A NOTIFICATION m ARRIVE FROM p_J TO p_0, BEFORE THE DELIVERING p_0 FOLLOW THIS TWO CHECKS...

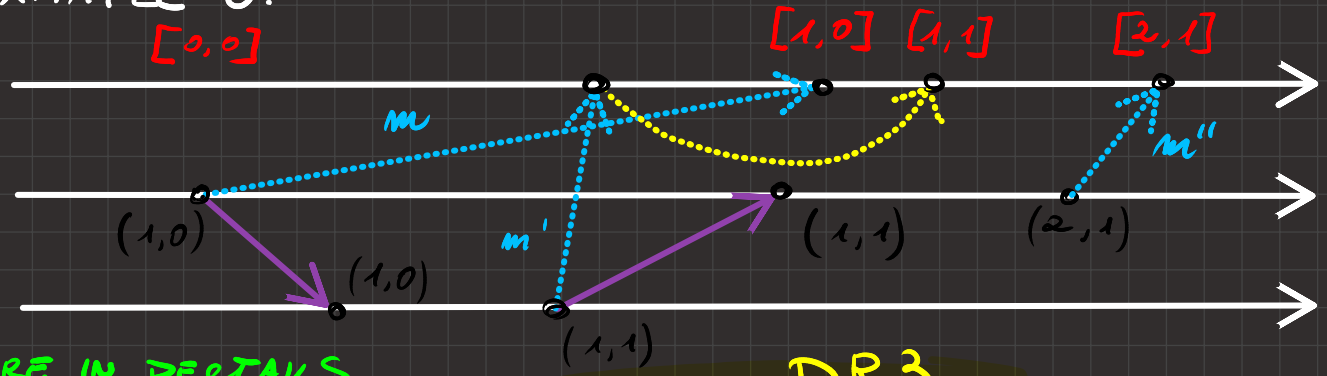
1. $D[J] = TS(m)[J] - 1$
2. $D[k] \geq TS(m)[k] \quad \forall k \neq J$

FOLLOWING THIS TWO RULES THE MONITOR IS SURE TO DELIVER THE MESSAGE IN CORRECT

ORDER.

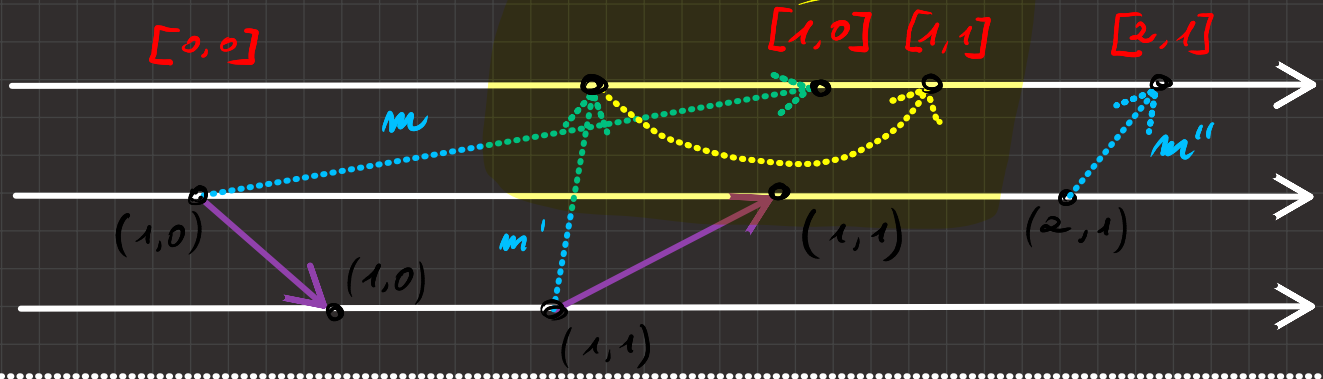
IF THE ANSWER TO THIS TWO POINTS IS NO THE MESSAGE IS PARKED IN A WAIT QUEUE

EXAMPLE 6.



MORE IN DETAILS

DR3



- THE MESSAGE m ARRIVES AFTER m' TO P_0 BUT $m \to m'$

$TS(m') = (1,1)$, $TS(m) = (1,0)$

THE CONDITION 2. IS NOT VERIFIED BECAUSE

$D = [0,0]$ AND $TS(m') = (1,1)$

$D[k] \geq TS(m)[k] \quad \forall k \neq j$ BUT $0 < 1$.

m' IS PARKED IN THE WAITING QUEUE AND DELIVERED WHEN THE CONDITIONS 1. AND 2. WILL B SATISFIED.

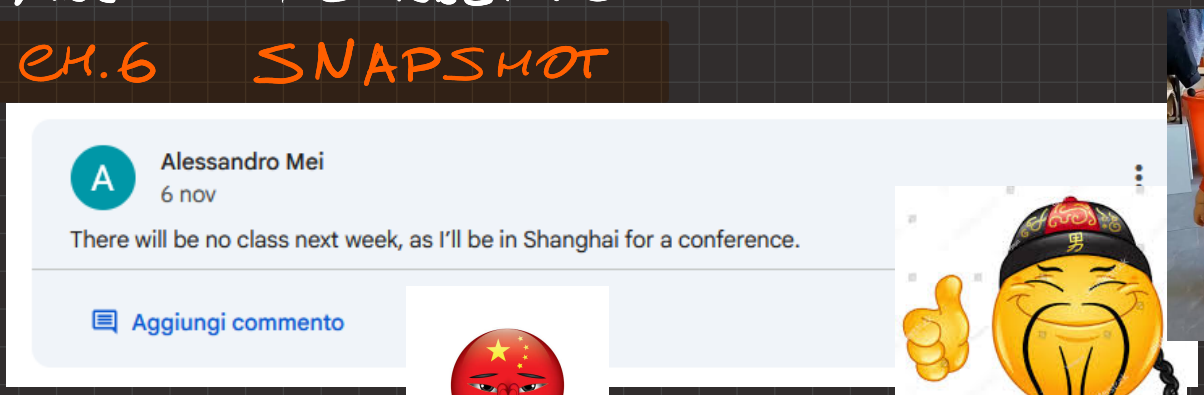
- HIDDEN CHANNELS

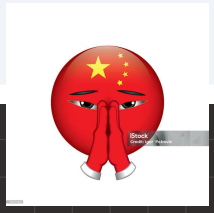
THIS WORKS VERY WELL IN SYSTEMS THAT DOESN'T COMPREEND PHYSICAL SYSTEMS.

- FOR EXAMPLE IF P_1 DETECT A BROCKEN TUBE AND P_2 DETECT LOW PRESSURE CAUSATED BY THE BROCKEN TUBE...

MESSAGES CAN ARRIVE INVERTED IN ORDER AND THE MONITOR CAN HAVE A DISTORTED PERCEPTION BECAUSE FOR IT THE MESSAGES ARE CONCURRENTS.

CH.6 SNAPSHOT











Corso Da Roma NOV. 9. 2026

IN THE CH2 THE MONITOR ASKED TO PROCESSES THEIR STATE, BUT THE GLOBAL STATE THAT COME OUT IS INCONSISTENT AND IT REQUIRED TO STOP COMPUTATION TO ASK AND RECEIVE STATES FROM PROCESSES...

- CANDY-LAMPORT
  - WHAT IS A SNAPSHOT ?

    A SNAPSHOT IS THE GLOBAL STATE AND MESSAGES ON CHANNELS OF A SYSTEM
  - WHAT IS CANDY LANDPORT ?
  - THAT IS AN ALGORITHM TO BUILD A CONSISTENT SNAPSHOT WITHOUT STOPPING THE COMUTATION.
  - IT USES A SPECIAL MESSAGE, THE MARKER MESSAGE. IT USES FIFO CHANNELS.

THE ALGORITHM:

1. P_0 SEND A MARKER TO HIMSELF

- WHEN p_i RECEIVE A MARKER FROM p_s

1. IF IT HAS NEVER RECEIVED A MARKER BEFORE
   1.1. SAVE ITS STATE
   1.2 SEND BROADCAST MARKERS
   1.3 MARK AS EMPTY THE COMMUNICATION CHANNEL $p_i - p_s$ AND STARTS LISTENING THE OTHER CHANNELS.
2. IF IT HAS ALREDY RECEIVED A MARKER BEFORE
   2.1 IT STOP COMMUNICATION $p_i - p_s$
3. IF IT HAS ALREDY RECEIVED MARKER FROM EVERYONE PROCESS IN THE D.S. IT SEND STATE AND MESSAGE QUEUE TO THE MONITOR.

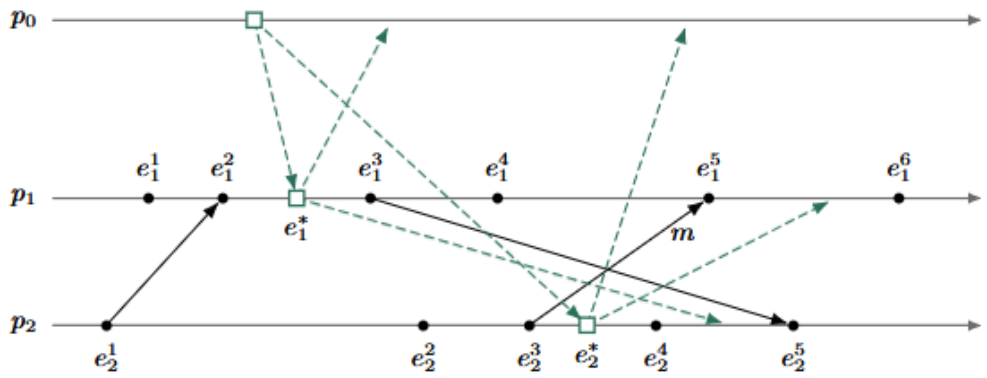
- MESSAGE COUNTING AND CONSISTENCY

MESSAGES CAN BE

1. SENDED BEFORE SENDER PHOTO AND RECEIVED BEFORE RECIPIENT PHOTO
   - COUNTED IN RECIPIENT LOCAL STATE
2. SENDED BEFORE SENDER PHOTO AND RECEIVED AFTER RECIPIENT PHOTO
   - COUNTED IN CHANNEL
3. SENDED AFTER SENDER PHOTO
   - NOT IN SNAPSHOT

- THAT GUARANTEES THAT THE SNAPSHOT IS CONSISTENT BECAUSE IS IMPOSSIBLE TO FIND A MESSAGE RECEIVED WITHOUT THE RELATIVE SEND EVENT.
- THAT SNAPSHOT CAN DESCRIBE A SITUATION THAT NEVER APPENED BUT IT IS CONSISTENT.

EXAMPLE. 7



- The message from e_2^1 arrives at p_1 before p_1's snapshot, like A in the second execution: it is already counted in σ_1^2;
- m leaves p_2 before p_2's snapshot, and arrives at p_1 after p_1's snapshot and before p_2's marker, like A in the first execution: it enters the list, and $\chi_{2,1} = \{m\}$;
- The message from e_1^3 leaves p_1 after p_1's snapshot, like B: it arrives at p_2 after p_1's marker, when the list is already closed, and $\chi_{1,2} = \emptyset$.

The snapshot is the global state Σ^{23} (two events of p_1, three of p_2), with m in transit from p_2 to p_1.

Exercises 2

Alessandro Mei • 2 ott

Exercise 6: Show that the history of an event is a consistent cut.

Exercise 7: Show that the Chandy-Lamport Snapshot Protocol builds a consistent global state.

Exercise 8: Show that the Chandy-Lamport Snapshot Protocol can build a global state that never happened.

Exercise 9: What good is a distributed snapshot when the system was never in the state represented by the distributed snapshot? Give an application of distributed snapshots.

Exercise 10: Consider a distributed system where every node has its physical clock and all physical clocks are perfectly synchronized. Give an algorithm to record global state assuming the communication network is reliable. (Note that your algorithm should be simpler than the Chandy–Lamport algorithm.)

Exercise 11: What modifications should be done to the Chandy–Lamport snapshot protocol so that it records a strongly consistent snapshot (i.e., all channel states are recorded empty).

Exercise 12: Show that, if channels are not FIFO, then Chandy–Lamport snapshot algorithm does not work.

Exercise 13: Let S0 be the global state when the Chandy-Lamport snapshot protocol starts, S be the global state built by the protocol, and S1 be the global state when the protocol ends. Show that S is reachable from S0 and that S1 is reachable from S. Remember that S might not have happened. (In other words, there is a consistent run that goes through S0, S1 and S.)

EXERCISE 6:

SHOW THAT THE HISTORY OF AN EVENT IS A CONSISTENT CUT

mmmh ...

1 A CUT IS A SUBSET OF THE GLOBAL HISTORY

2 A CONSISTENT CUT C IS A CUT WHERE IF $e, e' \in H, e' \in C \Rightarrow e \rightarrow e'$

3 THE HISTORY OF AN EVENT e IS DEFINED AS:

$$\phi e = \{e' : e' \to e\} \cup \{e\}.$$

LET'S SUPPOSE THAT (ϕe) IS NOT A CONSISTENT CUT...

4 IF $e' \in (\phi e)$

$e' \to e$

SUPPOSE THAT:

$\exists\, e'' \notin (\phi e)$ $e'' \to e'$ (BUT) FOR DEFINITION OF (ϕe)

$e'' \in (\phi e)$

THAN, THE CUT IS CONSISTENT

EXERCISE 7

DEMONSTRATE THAT CANDY-LANDPORT BUILDS CONSISTENT GLOBAL STATE...

- WE CAN DIVIDE CANDY-LANDPORT IN TWO STAGES
  1. PREREGISTRATION
  2. POSTREGISTRATION
- WHEN A PROCESS TAKE THE PHOTO TO ITS STATE IT IS THE POSTREGISTRATION PHASE, BEFORE THAT, IT IS IN PREREGISTRATION PHASE.
- WE CAN FIND 3 TYPE OF MESSAGE SITUATIONS
  1. SENDER SEND A MESSAGE BEFORE THE REGISTRATION AND THE RECIPIER RECEIVE THE MESSAGE BEFORE REGISTRATION

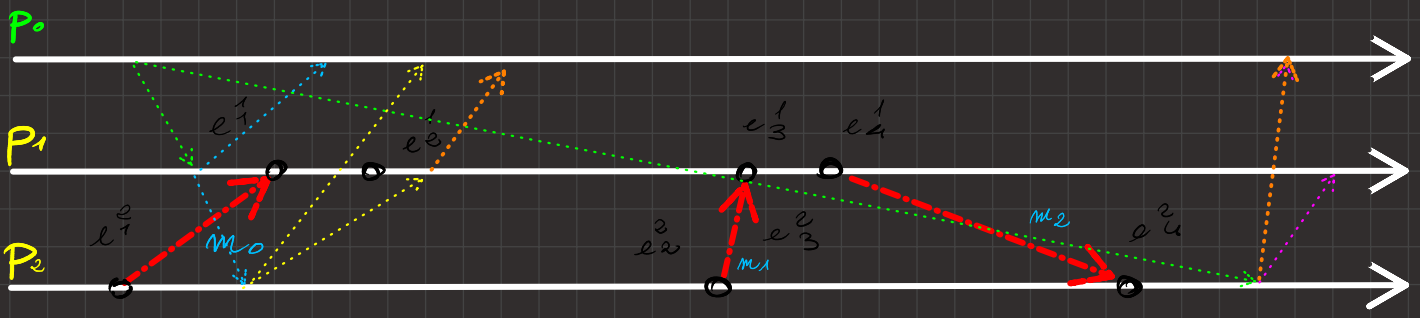
2. SENDER SEND A MESSAGE BEFORE THE REGISTRATION AND THE RECIPIER AFTER THE MESSAGE BEFORE REGISTRATION
3. SENDER SEND A MESSAGE AFTER THE REGISTRATION

- IN CASE 1. THE SEND ACTION IS REGISTERED IN THE LOCAL STATE OF THE SENDER AND THE RECEIVE ACTION IS REGISTERED IN THE LOCAL STATE OF THE RECIPIER.
- IN CASE 2. THE MESSAGE IS CONSIDERED ON CHANNEL
- IN CASE 3. THE MESSAGE IS NOT IN THE SNAPSHOT.
- WITH THIS ARCHITECTURE IS IMPOSSIBLE TO FIND EVENTS WITHOUT ITS RELATIVE CAUSE IN THE BUILT STATE.

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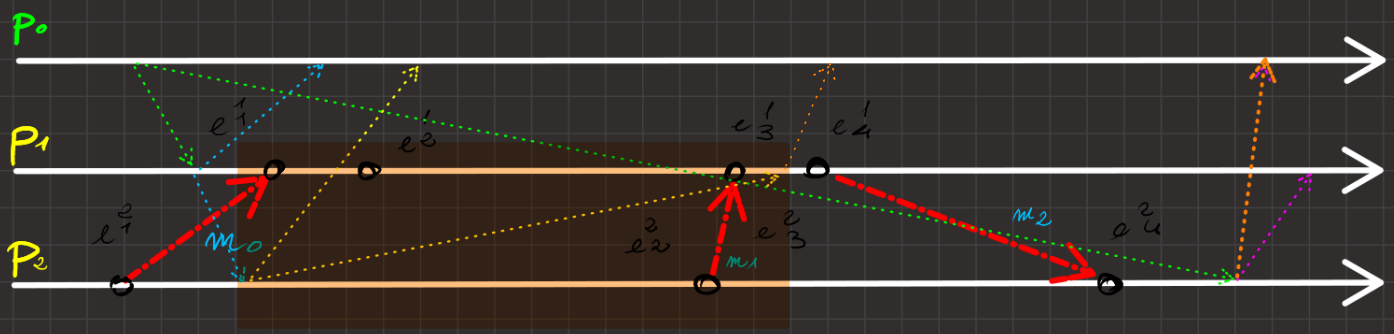
EXERCISE 8

DEMONSTRATE THAT CANDY-LANDPORT CAN BUILD GLOBAL STATES THAT NEVER HAPPENED...



$$SNAPSHOT = \{\{e_1^2\}, \{m_0\}\}$$

IS A CHRONOLOGICAL IMPOSSIBLE GLOBAL STATE.



SITUATION LIKE THIS ARE IMPOSSIBLE BECAUSE KANDY LANDPORT USES FIFO CHANNELS.

$SEND_i(m_0) \rightarrow SEND_i(m_1) \Rightarrow DELIVER_j(m_0) \rightarrow DELIVER_j(m_1)$

ESERCIZIO 9

GIVE AN EXAMPLE OF APPLICATION OF DISTRIBUITED SNAPSHOTS.

- FOR EXAMPLE IF YOU WANT TO VERIFY IF A PROCESS IS NOT CRASHED
- ANY STABLE PROPERTIES

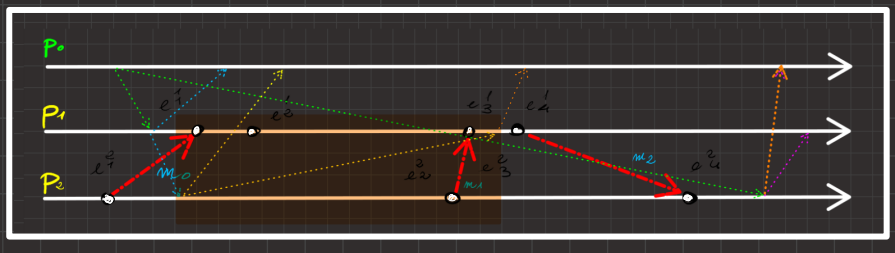
ESERCIZIO 10

1. P_0 SENDS "START" MESSAGE TO ALL THE PROCESSES
2. P_i RECIEVE "START", SAVE ITS STATE AND LISTEN THE IN CHANNELS WHILE t IS LESS THAN A T
3. P_i SENDS STATE AN STATE OF CHANNELS TO P_0.

ESERCIZIO 11

STRONGLY CONSISTENT ?

ESERCIZIO 12



IN THIS CASE WE HAVE m_1 IN THE SNAPSHOT BUT THE EXPEDITION OF m_1 HAPPENED A LOT OF TIME AFTER THE MOMENT CAPTURED IN THE SNAPSHOT.

Exercise 13: Let S0 be the global state when the Chandy-Lamport snapshot protocol starts, S be the global state built by the protocol, and S1 be the global state when the protocol ends. Show that S is reachable from S0 and that S1 is reachable from S. Remember that S might not have happened. (In other words, there is a consistent run that goes through S0, S1 and S.)

WE HAVE TO DEMONSTRATE THAT:

$\Sigma_{S0} \rightarrow \Sigma_S \rightarrow \Sigma_{S1}$

- WE KNOW THAT GIVEN $c_1 \subseteq c_2$ WHERE c_1 AND c_2 ARE TWO CONSISTENT CUT, EXIST A PATH $c_1 \rightarrow c_2$ ON THE LATTICE OF STATES...
- $\Sigma_{S0} \subseteq \Sigma_S \subseteq \Sigma_{S1}$ AND ALL THIS CUT ARE CONSISTENT

THEN

EXIST A PATH $\Sigma_{S0} \rightarrow \Sigma_S \rightarrow \Sigma_{S1}$. ☑

CH. 7. FAILURE, ATOMIC COMMIT AND TPC

UNTIL NOW SYSTEMS DOESN'T CRASH AT ALL BUT REAL SYSTEMS CRASH AND WE HAVE TO HANDLE THESE PROBLEMS...

DISTRIBUITED TRANSACTIONS

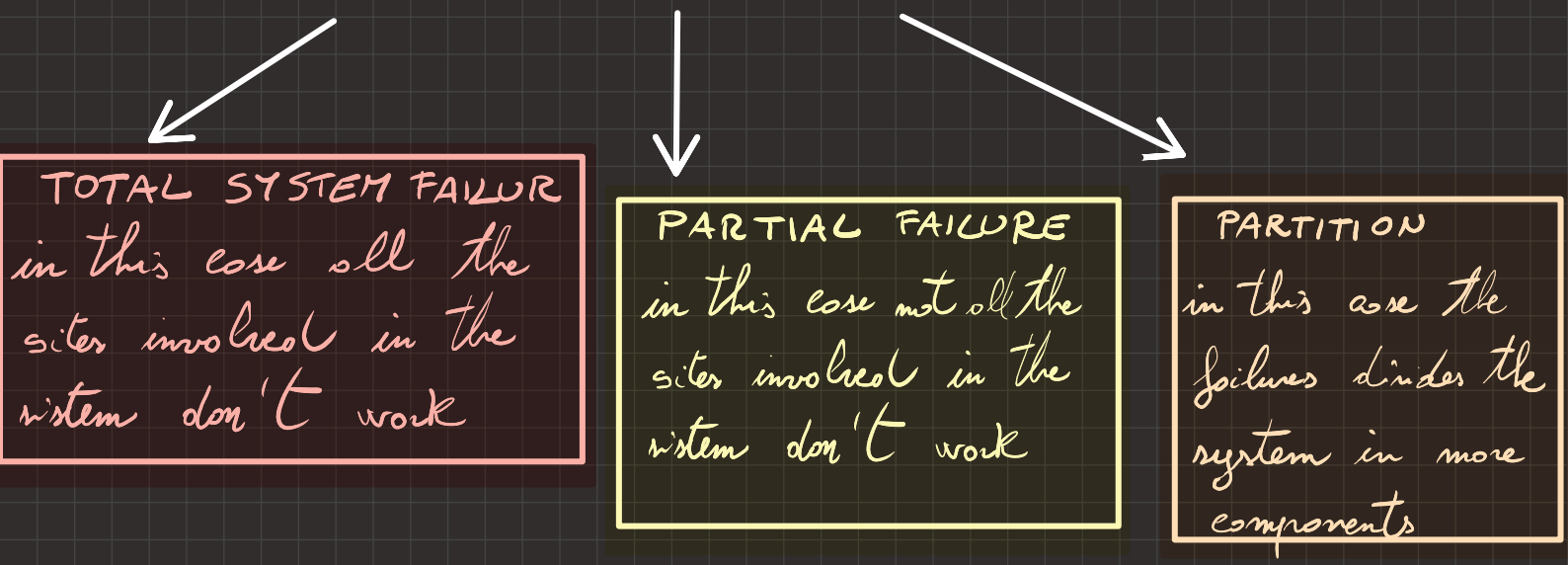
Carlo De Rome NOV. 10. 2026

- IN A DISTRIBUTED SYSTEM WE CAN IDENTIFY SOME PARTS

1. SITES = PROCESSES
2. HOME SITE = A SYSTEM THAT HANDLE THE OPERATIONS OF ONE SPECIFIC SITE
3. TRANSACTION MANAGER = A SYSTEM IN THE HOME SITE THAT HANDLE WRITING AND LECTURES ON THE SITE.

- WHEN A SITE WANT TO DO A DISTRIBUTED TRANSACTION ALL THE OTHER SITES HAVE TO ACCEPT THE TRANSACTION OTHERWISE THE TRANSACTION TERMINATES WITH AN ABORT.

- THE ABORTION OF A TRANSACTION IS NOT THE MOST IMPORTANT PROBLEM IS THE CONSISTENT TERMINATION... WE WANT THAT ALL THE SITES DO THE SAME OPERATION FOR THE TRANSACTION ABORT OR COMMIT
- A SYSTEM FAILUR COULD BE



- THE SITES WORKS FOLLOWING THE FAIL STOP PROTOCOLL, A SITE CAN ONLY ⇄ WORK / DON'T WORK BUT IT CAN'T WORK PARTIALLY.
- BEFORE DOING A DISTRIBUTE TRANSATION A COORDINATOR SITE HAS TO CHECK IF 2. RULES ARE RESPECTED BY ALL THE PARTECIPANT SITES.
  1. THERE ARE N'T NOT-CONFIRMED TRANSACTIONS
  2. THERE AREN'T DATA ON THE VOLATILE MEMORY OF THE SITE.
- ALL THE SITES HAVE VOLATICE MEMORY (DT LOG) WHERE THEY COULD SAVE DATA USEFULL FOR RECOVERY.
- WHEN A TRANSACTION ARRIVE TO THE SITE, THE SITE CAN ANSWER WITH YES OR NO... IF EVEN JUST A SITE ANSWER NO THE COMMON DECISION IS ABORT...



NO: 1
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- ATOMIC COMMIT PROTOCOL

EACH SITE CAN ANSWER WITH YES OR NO ... IF EVEN JUST A SITE ANSWER NO THE COMMON DECISION IS ABORT...

AN ATOMIC COMMIT PROTOCOL IS AN ALGORITHM THAT MAKE A DECISION FOLLOWING THESE 5 RULES:

1. ALL THE PROCESSES THAT REACH A DECISION REACH THE SAME
2. A PROCESS CAN'T CHANGE ITS DECISION
3. THE COMMIT IS REACHED IF ALL THE SITES VOTED YES
4. IF THERE AREN'T FAILURES AND ALL THE SITES VOTED YES, THE DECISION IS COMMIT
5. IN EACH EXECUTION IF SOME FAILURES ARE SOLVED AND THEY DON'T OCCURS FOR A LONG TIME SOONER OR LATER THE SISTEM WILL TAKES A DECISION.

- THE PROCESSES COULD BE IN AN UNCERTAIN PERIOD AND IT COULD CRASH IN THIS UNCERTAIN PERIOD...

I SCENARIO : IT CANNOT COMUNICATE WITH OTHER PROCESSES AND IT CANNOT RETRIVE INFORMATIONS BECAUSE IT IS BLOCKED IN A COMPONENT OF ONE PARTITION

II SCENARIO : IT CAN COMMUNICATE WITH OTHER PROCESSES AND RETRIVE INFORMATIONS.

- TIMEOUTS

TO CHECK IF A SITE IS CRASHED WE CAN USE TIMEOUTS

BUT

IS DIFFICULT TO CHOSE A CORRECT TIME FOR THE TIMEOUT AND SOMETIMES IT COULD SET AS TIMEOUTED GOOD SITES.

TWO LIMITS OF ATOMIC COMMIT PROTOCOL
- IF FAILURES ARE POSSIBLE EACH ACP CAN BLOCK PROCESSES
- ACPs CAN'T GUARANTEE INDEPENDENT RECOVERY

• TWO-PHASE COMMIT PROTOCOL

2 THINGS BEFORE STARTING:

Yes: IS THE CHOICE OF A PROCESS

YES: IS A MESSAGE

yes: IS A DTLOG LOG



: FAMOUS MEME 2021-2023

• THE PROTOCOL

1. THE COORDINATOR SENDS TO ALL THE PARTECIPANTS A MESSAGE "VOTE REQ"
2. EACH PARTECIPANT CAN ANSWER WITH "YES" OR "NO" IF IT VOTE "NO" IT DECIDE Abort AND IT STOPS.
3. THE COORDINATOR PEERS ALL THE VOTES AND DECIDES commit AND SENDS COMMIT MESSAGE OR DECIDES Abort AND SENDS ABORT MESSAGE TO THE COORDINATOR.
4. EACH PARTECIPANTS WAIT FOR COMMIT OR ABORT MESSAGE.

• TERMINATION PROTOCOL

- THE PROCESSES COULD BE IN UNCERTAIN STATE. THE SYSTEM REQUIRE A SOLUTION TO AVOID INFINITE EXPECTATIONS.
- EACH SITE p HAS A TIMEOUT, IF IT EXPIRES THE SITE RUNS THE TERMINATION PROTOCOL:

1. p THAT IS THE INIZIATOR SENDS TO THE OTHERS (RESPONSERS) DECISION-REQ, IF IT DOESN'T RECIVE A RESPONSE BEFORE THE END OF THE SECOND TIMEOUT IT RE-TRY THE TERMINATION PROTOCOL.
2. THE RESPONSERS RECEIVE THE REQUEST AND:
   - 2.1 IF A RESPONSER DECIDED Abort OR commit IT SENDS ABORT OR COMMIT.
   - 2.2 IF IT DOESN'T ALREDY DECIDED, IT SEND ABORT MESSAGE
   - 2.3 IT VOTED YES BUT IT NOT ALREDY DECIDED, IT DON'T SEND N

• RECOVERY, DT LOG, GARBAGE COLLECTOR

- BEFORE EXECUTING ANY ACTION THE SITE WRITE THE ACTION IN ITS DTLOG, FOR EXAMPLE BEFORE VOTING YES TO A VOTE-REQ THE SITE WRITE "YES" ON THE DT LOG, THIS CAN BE ERASE IF ALL THE DATAS OF THE TRANSACTION ARE IN STABLE MEMORY AND IF THE TRANSACTION IS ENDED WITH ABORT OR COMMIT.
- IF A SITE CRASH IT CAN EASLY RETURN AT THE STATE BEFORE THE CRASH FOLLOWING THIS PROTOCOL.

RECOVERY PROTOCOL:

1. IF IT CRASH BEFORE SEND A MESSAGE OR WRITING ON THE DT LOG IT DECIDE ABORT
2. IF IT DECIDED SOMETHING CAN EASLY RESTORE ITS STATE
3. IF IT WAS IN UNCERTAIN STATE IT STARTS TERMINATION PROTOCOL.

• 2PC EVALUATION

- ROUND

TIME COMPLEXITY IS MEASURED IN ROUNDS.
A ROUND IS THE MAX TIME THAT A MESSAGE USE TO ARRIVE.

* TIME COMPLEXITY: WITHOUT CRASHES IT REQUIRES 3 ROUNDS. THE TERMINATION PROTOCOL ADDS 2 ROUNDS.
* SPACE COMPLEXITY: WITHOUT CRASHES IT USES $3m$ MESSAGES (WHERE m IS THE NUMBER OF PARTECIPANTS) IF ALL THE PARTECIPANTS GOES IN TIMEOUT THE MESSAGES BECOME $m(3m+1)/2$.

* IT IS RESILIENT, BECAUSE IT TOLLERATES ALL TYPE OF FAILURES AND BLOCKS
* SITES MIGHT BLOCK IN UNCERTAIN PHASE.

CH. 8. CONSENSUS AND PAXOS.